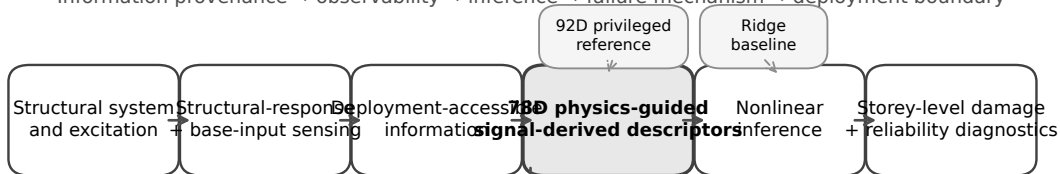


Deployment-aware structural damage inference

Information provenance → observability → inference → failure mechanism → deployment boundary



Base/ground-input sensor assumed available

When is damage inference identifiable and reliable under realistic sensing and information constraints?

Measurement-noise pathway

Structural-response measurement noise

Descriptor-space distribution shift

Prediction inflation and broadening

High-damage bias reversal

Upper-bound prediction saturation

Sensor-sparsity pathway

Structural sensor removal

Dependency-aware descriptor unavailability

Reduced cross-storey observability

Increasing severe-damage underprediction

Marginal value of additional sensing

Directional calibration helps in-distribution,

but becomes misaligned after bias reversal

15 layouts • 28 one-sensor additions

complete, tested four-sensor subset lattice

Distinct degradation mechanisms: noise changes error direction, sensor loss reduces observability